HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Repetition to

# ADVANCED TOPICS IN ENGINEERING PHYSICS

***'Plasma and its applications'***

***including answers***

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# 1 Basics of plasma physics

- 1. Give an approximate value for the electron density of the ionosphere.
  - About $10^{10}/m^3$

- 2. The colour of polar lights is usually either greenish or purple. Why?
  - The colour of polar lights directly depends on the composition of the magnetosphere, which, itself, depends on altitude. At an altitude of about 80 to 150 km polar lights are preferentially greenish due to the excitation and de-excitation of oxygen. At higher altitudes of approximately 150 to 600 km, nitrogen is the predominant element. This element emits red and blue radiation, resulting in purple light via additive mixture of colour.

- 3. Describe – in short – the $0^{th}$ and the $5^{th}$ state of aggregation.
  - $0^{th}$: matter at 0K (-273,15°C), aka Bose Einstein condensate
  - $5^{th}$: super plasma, where atomic nuclei decay

- 4. Are all plasmas fully ionised?
  - No. Also partially ionised gases are referred to as plasma.

- 5. What are positive and negative charge carriers within plasmas?
  - Positive: atoms and ions where at least one electron was removed (cations)
  - Negative: atoms and ions where at least one additional electron was attached (anions) or free electrons

- 6. What is the so-called ‘Crookes dark space’?
  - A dark zoned within the plasma discharge close to the cathode. Here, the number of charge carriers is quite low and the probability of excitation and subsequent de-excitation of atoms – and thus emission of light – is lowered.

- 7. What are cathode rays?
  - Cathode rays = rays of electrons emitted from the cathode material and moving from the cathode towards the anode

- 8. What are canal rays?
  - Canal rays = ion rays moving from the anode towards the cathode

- 9. Which interactions are dominant within plasmas, interactions between two charged particles or interactions between a charged particle and a neutral one…and why?
  - The interaction between two charged particles is considerably higher than with neutral ones…since the Coulomb range is much higher (by 5 orders of magnitude) than ranges of Van der Waals forces.

- 10. How many ions and electrons can be found within a gas at ambient conditions?
  - about $10^3/cm^3$

- 11. Thermal ionisation of atoms and dissociation of molecules starts at a temperature of…?
  - …about 1000K

- 12. What is a ‘non-equilibrium plasma’?
  - In non-equilibrium plasmas, the involved plasma species (electrons, ions…) feature different temperatures/energies/velocities.

- 13. Name at least three processes or effects that occur within plasmas.
  - Dissociation
  - Ionisation
  - Excitation (and accompanying emission of radiation)
  - Attachment

- 14. What is – roughly – the electron density of a so-called glow discharge?
  - about $10^{12}/m^3$-$10^{18}/m^3$ (see graph on page 56 of the handout)

- 15. Name at least two examples for a) plasma-chemical and b) plasma-physical mechanisms within plasmas.
  - a) reduction, oxidation
  - b) dissociation, ionisation, attachment, excitation

- 16. Explain shortly the function of a fluorescent lamp.
  - Ignition of plasma → emission of UV irradiation → conversion of UV light by fluorescent coating on tube

- 17. Do the electrodes of a glow discharge plasma source emit any electrons?
  - In a glow discharge, the electrodes remain ‘cold’ and do thus not emit any electrons via thermoemission, aka thermionic emission.

- 18. Above which temperatures does thermoemission, aka thermionic emission, occur?
  - about 10.000 to 30.000 K

- 19. Name at least three parameters that define the type or kind of a plasma discharge.
  - Pressure, gas (or gas mixture), electrode material, current density…

- 20. Name at least three characteristic properties of plasmas.
  - Quasi-neutrality, ambipolarity, Debye shielding, transport of particles via drift and diffusion…

- 21. Name at least three essential plasma parameters.
  - Temperatures and density of particles, Debye lengths and radii, degree of ionisation, plasma frequency…

- 22. Give a definition of ‘quasi-neutrality’.
  - On a macroscopic scale, a plasma contains an equal number of positive and negative charge carriers.

- 23. What is the difference between quasi-neutrality and strict neutrality?
  - Quasi-neutrality: permittivity $\varepsilon$ of plasma $<< 1$
    (in most plasmas valid for $\varepsilon < 10^{-2}$)
  - Strict neutrality, if electron density = sum of al products of ion charge and ion density

- 24. What is the underlying reason or effect for quasi-neutrality?
  - Electrostatic forces within plasma practically do not allow accumulation of charges

- 25. What is the so-called ambipolarity of a plasma?
  - For a two two-component plasma, just as much singly charged ions as electrons flow off per unit of time.

- 26. Shortly explain the effect of Debye shielding.
  - Accumulation of a few charge carriers in vicinity of a local charge due to electrostatic forces → shielding of charge

- 27. Give a definition for the Debye length.
  - Characteristic length (or radius for Debye spheres), where electrical potential of a local excess charge is reduced by a factor of 1/e (i.e., approximately 0.37). Within this range, quasi-neutrality is not fulfilled.

- 28. What is the so-called ‘sheath-effect’?
  - The condition of quasi-neutrality is not fulfilled within a thin plasma sheath at walls. In contrast to the undisturbed bulk plasma, a negative potential is found here. Hence, ions are accelerated towards the walls whereas electrons are reflected (plasma shielding).

- 29. In practice, what is a ‘wall’ in context with plasmas?
  - Any physical limit or boundary of a plasma, e.g. surfaces of electrodes, inner surfaces of vacuum or reactor chambers, sample or work piece surfaces etc.

- 30. What is the difference between ionisation and dissociation?
  - Ionisation = removal of an electron from its atom or molecule → cation (or vice versa → anion)
  - Dissociation = splitting of molecules, e.g. splitting of $O_2$ to 2O

- 31. Which effect is more probable, ionisation or dissociation?
  - Dissociation, since here, less energy is required (exmple: hydrogen, where dissociation energy = 4.5 eV and ionisation energy = 13.6 eV, i.e. a factor of 3)

- 32. What does the Boltzmann-equation express?
  - The Boltzmann-equation is a description of the statistic distribution of particles.

- 33. In which cases does the Boltzmann-equation apply?
  - For (i) long mean free lengths of path of the involved particles and (ii) collisions of merely two particles, i.e. two-body-collisions.

- 34. What is the mean free length of path?
  - = free distance, where no collision of propagating particle occurs

- 35. Name the pressure range of full vacuum.
  - $10^{-3}$ – $10^{-7}$ hPa

- 36. What is the key message of the drift-diffusion-approximation?
  - The motion or flux of charged particles within a plasma is described by both a drift component and a diffusion component.

- 37. Give a definition for the term ‘drift’.
  - ‘Drift’ is the motion of charged particles within and due to external fields.

- 38. The flux or current density of particles can be determined on the basis of which parameters?
  - Particle density, particle mobility, and electric field strength

- 39. What is diffusion of particles…and what is the underlying reason?
  - Diffusion is the compensation of different concentrations of particles. It is thus caused by concentration gradients.

- 40. What is a plasma in thermodynamic equilibrium?
  - In a plasma in thermodynamic equilibrium, all plasma species (ions, electrons, neutrals) feature the same energy/temperature.

- 41. Which equation is used for the determination of the degree of ionisation?
  - The Saha-equation.

- 42. Which plasma parameter has the major impact on the degree of ionisation?
  - Temperature

- 43. Give a short definition of the plasma frequency.
  - Plasma frequency = frequency of periodic oscillation of charge carriers within a plasma.

- 44. What is the difference between a (i) low temperature non-equilibrium plasma and a (ii) low temperature plasma in local thermodynamic equilibrium?
  - For (i), the electron temperature is much higher than the ion or gas temperature.
  - In case of (ii), all species feature a common temperature. For both plasmas, the gas temperature is comparatively low.

- 45. What is a so-called relativistic plasma?
  - In relativistic plasmas, the thermal energy is $E_{therm} \geq m_e \cdot c^2$ (i.e., about $10^6$ eV or higher).